Additional Post-Hoc Noise Robustness Analysis
Hardware alerts at the original frozen thresholds

Noise	Coverage	Model	Alert recall	Alert F1	Combined FPR
Clean	100.0000%	RF E4	0.9955	0.9896	0.0832%
Clean	100.0000%	LightGBM E4	0.9969	0.9902	0.0839%
1%	92.1941 ± 0.0737%	RF E4	0.9930 ± 0.0007	0.9882 ± 0.0004	0.0843 ± 0.0022%
1%	92.1941 ± 0.0737%	LightGBM E4	0.9944 ± 0.0012	0.9891 ± 0.0006	0.0821 ± 0.0026%
2%	84.4133 ± 0.0997%	RF E4	0.9730 ± 0.0015	0.9778 ± 0.0009	0.0866 ± 0.0035%
2%	84.4133 ± 0.0997%	LightGBM E4	0.9740 ± 0.0017	0.9778 ± 0.0010	0.0920 ± 0.0045%
5%	74.7809 ± 0.1046%	RF E4	0.8867 ± 0.0040	0.9192 ± 0.0020	0.2144 ± 0.0107%
5%	74.7809 ± 0.1046%	LightGBM E4	0.8889 ± 0.0026	0.8875 ± 0.0016	0.5775 ± 0.0163%
10%	69.5123 ± 0.1124%	RF E4	0.7780 ± 0.0040	0.7675 ± 0.0032	1.2591 ± 0.0278%
10%	69.5123 ± 0.1124%	LightGBM E4	0.8001 ± 0.0055	0.6770 ± 0.0040	2.8472 ± 0.0372%

Coverage is identical for RF and LightGBM; one shared column. Noisy values: mean ± sample SD across 10 paired seeds.

Coverage decreases because the unchanged input-quality rules can convert noise-induced negative electrical readings to missing values and trigger abstention.

End-to-end pipeline robustness under Gaussian sensor noise, including existing input-quality/abstention rules.